YONGFEI (FIONA) SHE

Software Engineering | C++ / Python / Java

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EDUCATION

University of Southern California **Expected May 2027**
Master of Science in Computer Science | Jan 2026 - Present

Chengdu University of Information Technology **Jun 2024**
Bachelor of Engineering in Computer Science and Technology

HONORS & AWARDS

ICPC: Invitational Silver (2023), Bronze (2022); CCPC**:** Women's Finals Bronze (2022).
National Scholarship**:** 1st in program (2023). University Merit Scholarship**:** Highest Tier (2021 & 2022).

TECHNICAL SKILLS

Languages: C++, Python, Java, C#, SQL, Lua
Engineering: Data Structures & Algorithms, Multithreading, ECS, Performance Optimization, Benchmarking
Tools: Git, Perforce, Visual Studio, CMake, .NET, Unreal Insights, RenderDoc

INDUSTRY EXPERIENCE

Software Engineer (C++ / Unreal Systems) **Sep 2023 - Dec 2025**
Chengdu Digital Sky Technology— Mandate Order (Steam) | Chengdu, China

- Cut connectivity computation by **85%** at a 1,000-unit workload with **spatial hashing** on a uniform grid; built templated KD-tree / union-find / DBSCAN for **5,000+ unit** scenes..
- Owned routing for **1,000+ concurrent agents**: heap-based **multi-source Dijkstra** on a dynamic road graph for threat-aware and supply routes, with batched road-state sync from GameWorld to AIWorld through a command queue..
- Owned the **data-driven** skill system for **100+ abilities** on the project's **multithreaded ECS**, decoupling event triggers from execution so new abilities are added through configuration, not code.
- Designed reusable **input and UI frameworks**: prioritized input contexts with reference-counted action monitoring, a stack-based UI layer manager, and a registry-based settings system.

RESEARCH EXPERIENCE

Research Assistant **Jun 2026 - Present**
UCLA, Artificial Intelligence & Visual Computing Lab

- **Co-first author** of *ArticulateArena: A Metric for Articulated Kinematics* (arXiv:2609.33931, 2026; submitted to **ICLR**).
- Contributed to a representation-invariant joint-motion metric and standardized evaluation protocols for generated articulated assets.
- Set up and ran multi-model experiments. Assessed asset quality and compared outputs to select assets suitable for downstream training.

SELECTED PROJECTS

ANGLE Graphics SDK & Developer Tools

- Built a .NET SDK on ANGLE with a native C++ bridge and dynamically loaded GLES bindings for offscreen and window rendering.
- Designed context-bound resource APIs and thread-access checks; generated bindings from Khronos XML with **automated signature and idempotency tests**.
- Added plugin/shader hot reload with failure handling, native/managed image comparisons, and rendering microbenchmarks.

AI-Assisted Shader Tool & Engine Subsystems **Jan - May 2026**

- Built an AI-assisted HLSL authoring tool with manual controls and editable template-based output; added **live recompilation** with last-working-shader fallback.
- Extended the provided PrimeEngine framework with multithreaded ECS simulation and independent update cycles.

C++ Software Renderer **Nov - Dec 2025**

- Wrote a CPU rasterizer from scratch with the C++ Standard Library, including depth buffering, perspective-correct interpolation, and programmable shaders.
- Optimized depth testing and fragment shading to process **5M+ fragments/sec** and render meshes with 10,000+ vertices.